Emergency Route System ko Google Maps se Alag Kaise Banayein

ML kahan use hota hai, aur extra ML features - Uncertainty-Aware Emergency Route and ETA Prediction System

1. Abhi ML kahan use ho raha hai

- **Segment travel-time prediction (XGBoost / LightGBM):** ek regression model har road segment ka time predict karta hai. Yehi A* ka edge weight banta hai.
- **Uncertainty prediction (quantile regression):** ek number ki jagah model lower guess (10th percentile) aur upper guess (90th percentile) deta hai. User ko range dikhti hai, jaise 12-18 min.
- **Fastest vs reliable route:** fastest route mein A* typical time pe chalta hai. Reliable route mein A* 90th-percentile (worst-case) time pe chalta hai.
- Pathfinding (A*), rerouting aur graph banana classical algorithms hain, ML nahi.

2. Quantile regression simple words mein

- Normal model ek hi number deta hai, jaise 15 minute. Real time kabhi exact nahi hota, isliye quantile regression range deta hai.
- Lower guess ka matlab: 10 mein sirf 1 trip isse fast hoti hai. Upper guess ka matlab: 10 mein sirf 1 trip isse slow hoti hai. 10 mein se lagbhag 8 trips dono ke beech aati hain.
- Example: ek road ka past time 10, 12, 13, 15, 15, 16, 18, 25 sec hai. Normal model bolega 15. Lower quantile bolega 11. Upper quantile bolega 22.
- Dono models ka data aur features same hote hain. Sirf training ka loss (target) alag hota hai.

3. Features jo Google Maps nahi deta

- **Ambulance-specific travel time (sabse strong idea):** ambulance siren bajati hai, signal todti hai aur traffic usko jagah deta hai. Asli ambulance trips pe train karo, ya ek correction model banao jo car ke time ko ambulance ke time mein badle.
- **Nearest nahi, best hospital:** har hospital ka ER waiting time predict karo (beds, specialty, bheed). Wo hospital chuno jahan total time = travel time + wait time sabse kam ho.
- **Emergency type ke hisaab se route:** heart ya trauma case mein smooth road chahiye. Road quality, speed breakers, potholes aur turns ki ginti ko A* ka extra objective banao.
- **Traffic signal aur junction ka delay:** time of day, junction type aur turn direction se har junction ka waiting time predict karo.
- **Aage ke jam ki prediction (proactive rerouting):** predict karo ki '10 min baad is road pe jam ka 70% chance hai', aur jam hone se pehle reroute karo.
- **Weather aur events:** baarish, kohra, school timings, match, festival, procession. SHAP ya feature importance se dikhao ki ETA kyun badha.
- **Confidence ke saath decision:** Google ek number deta hai. Ye project range aur confidence deta hai, aur range bahut badi ho to safer route suggest karta hai.

4. Final-year project ke liye recommendation

- Ye 3 lo, kyunki inme ML sach mein kaam karta hai: (1) hospital total time = travel + wait, (2) ambulance correction model, (3) proactive rerouting ke liye aage ki traffic prediction.
- Road quality aur signal delay baad mein add karo, agar time bache.

5. Evaluation

- Central prediction ke liye MAE aur RMSE report karo.
- Coverage bhi report karo: asli time kitni baar predicted range ke andar aaya. 10th-90th range lagbhag 80% trips cover karni chahiye.

6. Risks aur open points

- **Data sabse bada risk hai.** Hospital wait time aur asli ambulance trips public mein mushkil se milte hain. Jahan data nahi hai wahan synthetic data use karo aur report mein saaf likho.
- Har segment ke quantiles jodne se route ki range zyada badi dikhti hai. Student project mein jodna theek hai, bas limitation likh do.
- Open decisions: 'reliable' ki definition (worst-case 90th percentile recommended hai), update frequency (30-60 s), alert threshold, aur A* heuristic (straight-line distance / max speed).
- Ye document sirf do project PDFs ke basis pe hai. Koi dataset check nahi kiya gaya.